

Tree of Thoughts: Deliberate Problem Solving with Large Language Models

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Background

Standard LLMs are...

- autoregressive, left to right token predictors
- **BAD** at backtracking, exploration, fixing errors

Idea: Branching exploration (**tree**) improves final result

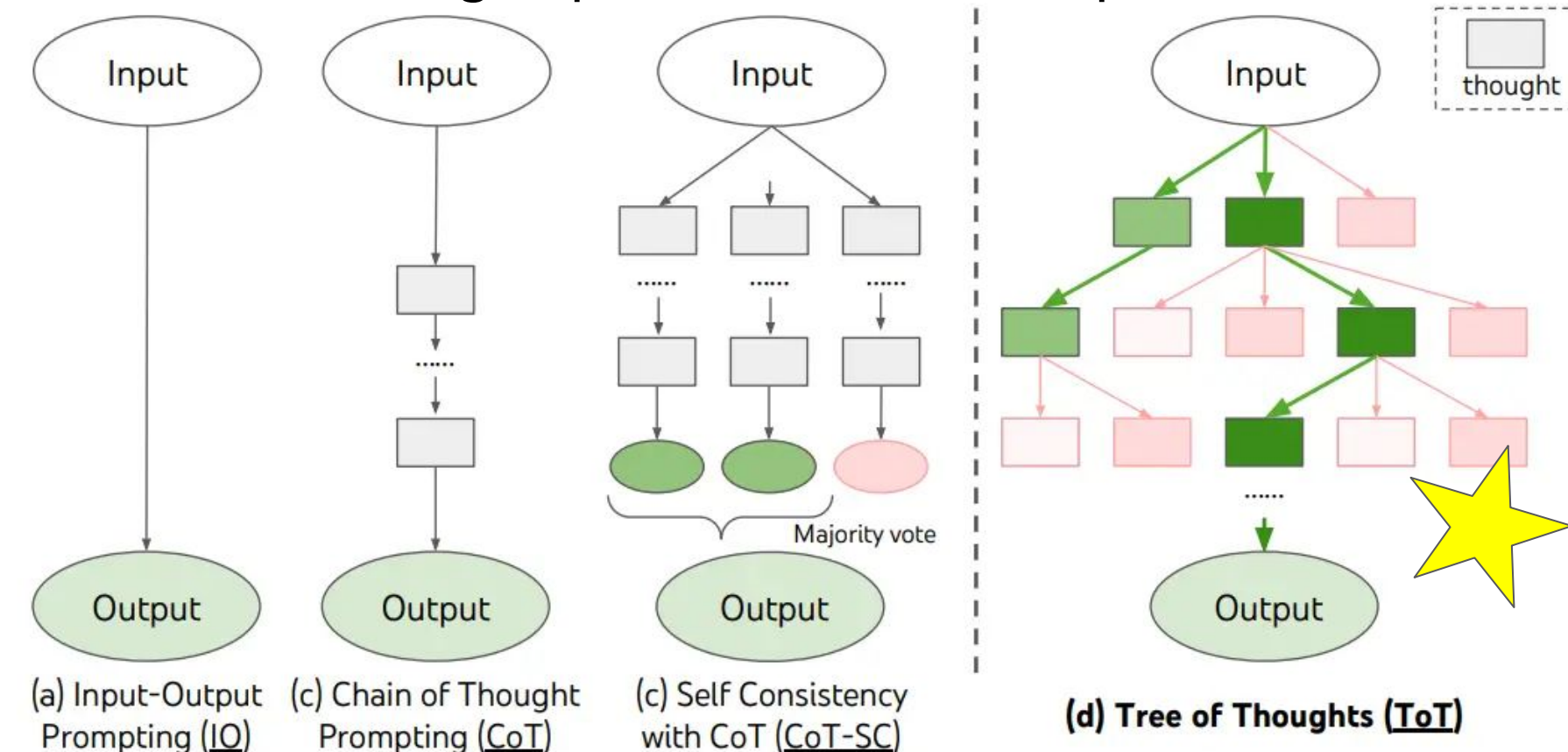


Fig 1. Yao et al. (2023) Prompting techniques using reasoning.

Paper Findings

Method	Success
IO prompt	7.3%
CoT prompt	4.0%
CoT-SC (k=100)	9.0%
ToT (ours) (b=1)	45%
ToT (ours) (b=5)	74%



4nums.com

- 100 difficult games
- mathematical reasoning

Table 2: Game of 24 Results.

randomwordgenerator.com

- 100 sentences
- creativity

goobix.com

- 156 5x5 boards
- problem solving

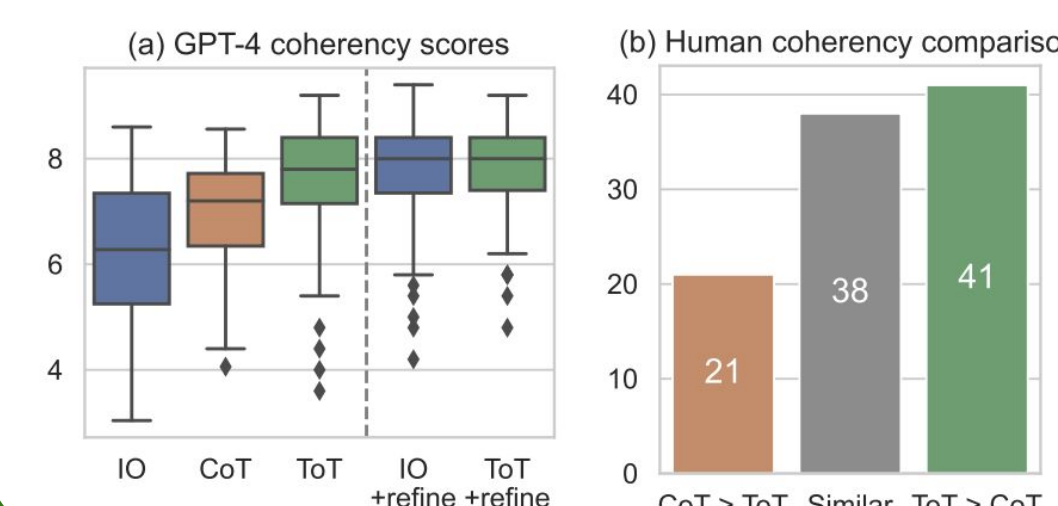
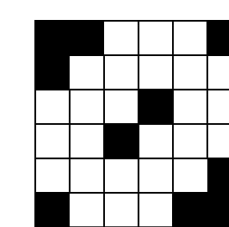


Figure 5: Creative Writing results.

Method	Success Rate (%)		
	Letter	Word	Game
IO	38.7	14	0
CoT	40.6	15.6	1
ToT (ours)	78	60	20
+best state	82.4	67.5	35
-prune	65.4	41.5	5
-backtrack	54.6	20	5

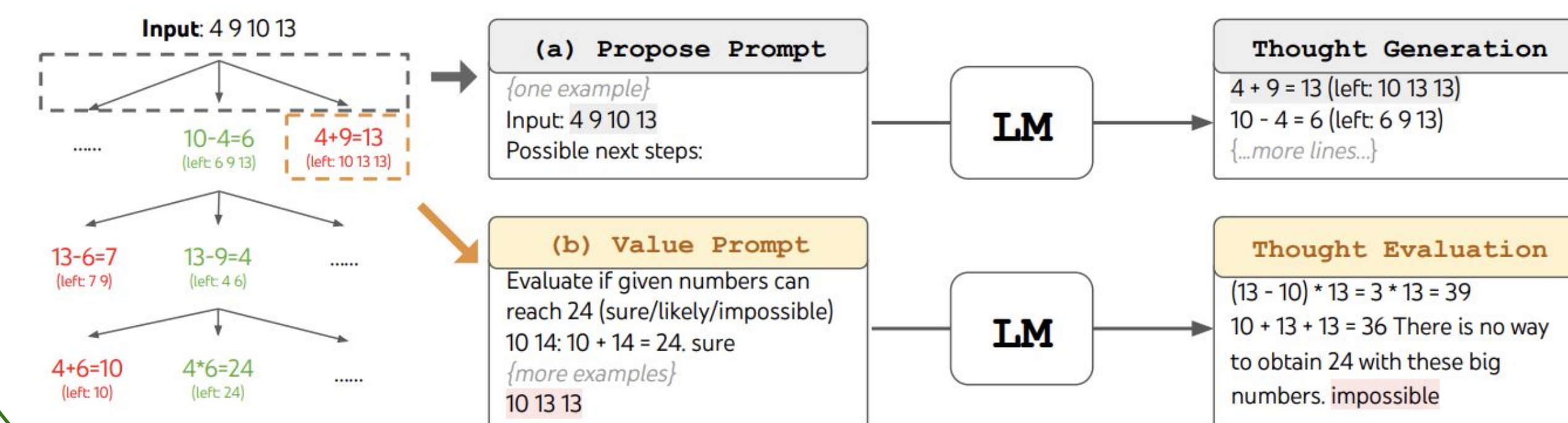
Table 3: Mini Crosswords results.

Methodology

1. Thought Decomposition
2. Thought Generation
3. State Evaluation
4. Search Algorithm

◆ Gemini 2.5 Flash

	Game of 24	Creative Writing	5x5 Crosswords
Input	4 numbers (4 9 10 13)	4 random sentences	10 clues (h1. presented;...)
Output	An equation to reach 24 (13-9)*(10-4)=24	A passage of 4 paragraphs ending in the 4 sentences	5x5 letters: SHOWN; WIRRA; AVAIL; ...
Thoughts	3 intermediate equations (13-9=4 (left 4,4,10); 10-4=6 (left 4,6); 4*6=24)	A short writing plan (1. Introduce a book that connects...)	Words to fill in for clues: (h1. shown; v5. naled; ...)
#ToT steps	3	1	5-10 (variable)



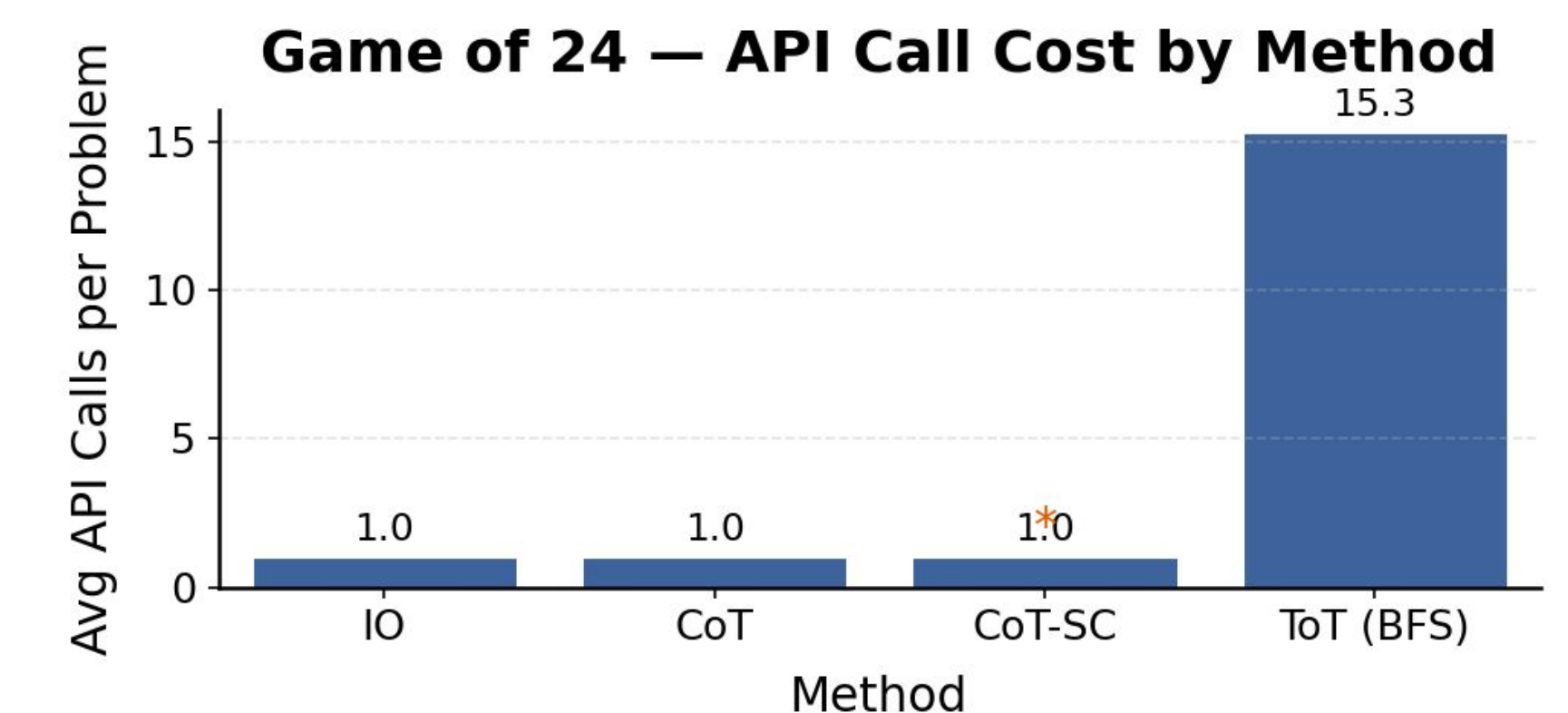
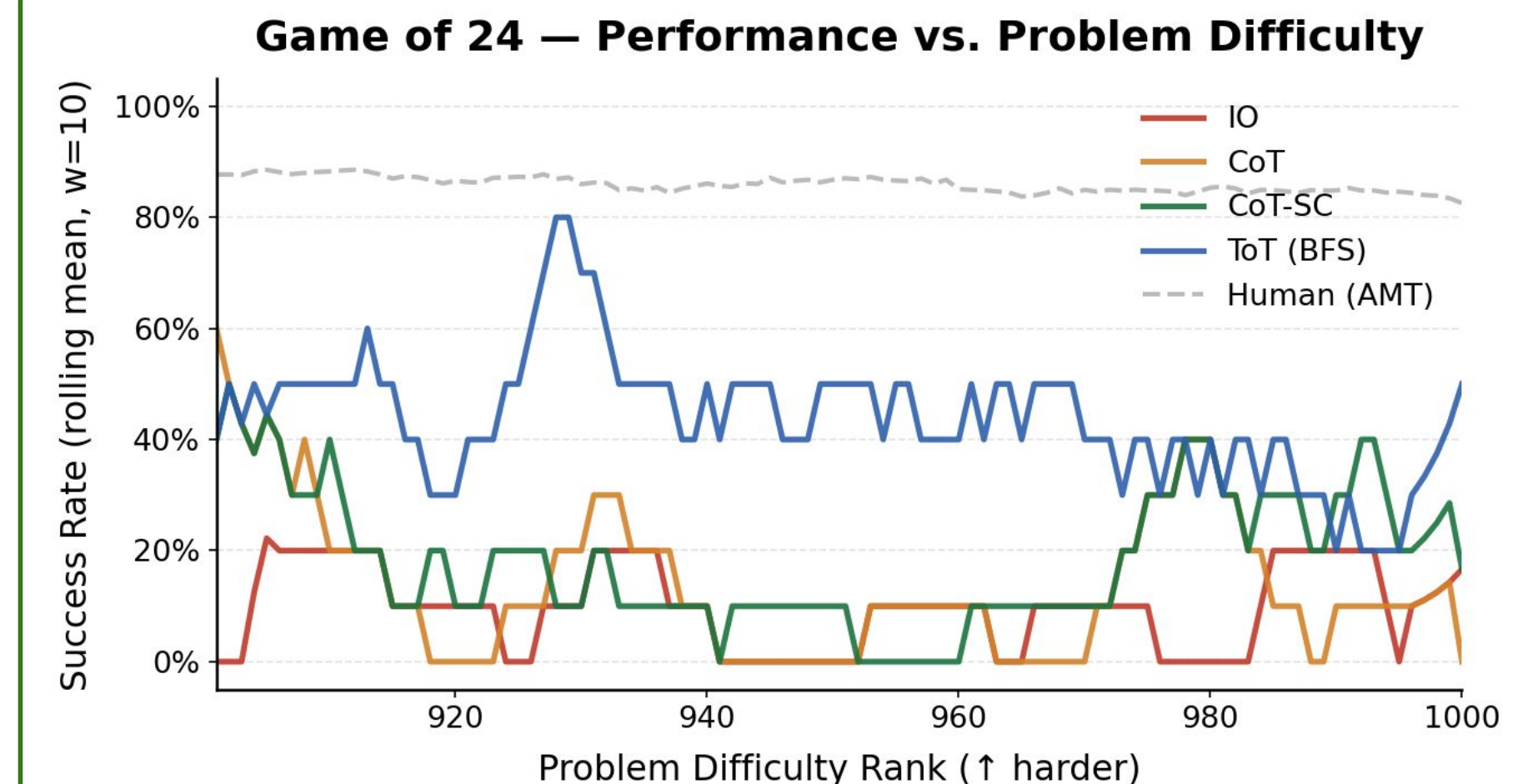
Conclusion / Future Work

- ToT **significantly** improves success rate in tasks that require multi-step reasoning
- ToT implementation **significantly** increases costly resource usage (i.e. LLM API calls)
- ToT reasoning is **weakly correlated** with human accuracy (+0.168, p = 0.094)
- Model reasoning is **brittle**, even with ToT and 3 “thoughts”

Future Work:

- Tuning the optimal number of search branches for a problem
- Utilizing mixed model types across a single problem to reduce resource requirements
- Analyze failure cases to determine cause of failure (e.g. arithmetic, heuristic errors)

Results



* CoT-SC true cost ≈ 5 calls/problem (Gemini batch counted as 1)

